

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
12	(a)	(i)		all four carbonates (1) acids react with metal carbonates to form carbon dioxide gas (1)	2			2		2
		(ii)		(hydrochloric acid would form) <u>insoluble</u> compound with Pb^{2+} (1) should use nitric acid / ethanoic acid (1)			2	2		2
		(iii)		award (2) for all four correct award (1) for any two correct $\text{Mg}^{2+}(\text{aq})$ white (precipitate) $\text{Fe}^{2+}(\text{aq})$ (dark) green (precipitate) $\text{Cr}^{3+}(\text{aq})$ (grey) green (precipitate) $\text{Pb}^{2+}(\text{aq})$ white (precipitate)	2			2		2
		(iv)		add excess sodium hydroxide (1) award (1) for either of following - magnesium hydroxide white precipitate remains but lead hydroxide precipitate dissolves (giving a colourless solution) - iron(II) hydroxide green precipitate remains but chromium hydroxide precipitate dissolves (giving a dark green solution) lead and chromium are amphoteric (iron and magnesium are not) (1)			3	3		3

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	(b)	(i)		$\text{Mg}^{2+} \quad \frac{91}{24.3} = 3.7449$ $\text{Ca}^{2+} \quad \frac{50}{40.1} = 1.2469 \quad (1)$ ratio is 3Mg: 1Ca (1) formula must contain $4 \times \text{CO}_3$ so it is $\text{Mg}_3\text{Ca}(\text{CO}_3)_4 \quad (1)$ accept alternative correct representations e.g. $\text{CaCO}_3.3\text{MgCO}_3$ $(\text{MgCO}_3)_3.\text{CaCO}_3$		1	2	3	2	
		(ii)		concentration = $1.25 \times 10^{-6} \text{ mol dm}^{-3}$ (from calcium or magnesium concentrations) (1) moles of solid used = $\frac{220 \times 10^{-6}}{353} = 0.6232 \times 10^{-6} \text{ mol} \quad (1)$ volume = $0.50 \text{ dm}^3 \quad (1)$ allow any value in the range 0.498-0.501 dm^3		1 1	1	3	2	
				Question 12 total	4	3	8	15	4	9